

Predicting Domain-Adaptation Learning Curves on PACS from Model–Data Relationships

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September 2026

Abstract

Estimating how target-domain performance changes with the amount of labeled target data usually requires adaptation runs at several data sizes. This pilot study illustrates an alternative forecasting framework on PACS: predicting a target-domain learning curve from information available before adaptation and refining that forecast after only limited early adaptation.

Each of 96 source–target cases is represented by

$$A(n) = A_0 + (a - A_0) [1 - (n + 1)^{-c}],$$

where n is the number of labeled target examples, A_0 is zero-shot test accuracy, a is an asymptotic parameter, and c controls the improvement rate. The fitted curves approximate the six measured test points with mean MAE 0.0287.

In cold start, metadata Ridge achieved an aggregate MAE of 0.0896 versus 0.1212 for the mean-curve baseline. In warm start, after observing the 5% and 10% adaptation stages, three-point logarithmic extrapolation achieved 0.0521 and metadata-plus-observation Ridge 0.0600. Global representation-discrepancy features did not improve the strongest predictors. Grouped ablations further showed that early validation performance was the most useful warm-start information family: removing it increased MAE from 0.0600 to 0.0827, whereas removing highly redundant budget or training-dynamics features slightly improved performance. The experiment therefore provides a compact example of how data-sufficiency forecasting could be studied within the proposed PhD direction.

1 Introduction

Domain adaptation transfers a model trained on one or more source domains to a related target domain with a different data distribution [PY10]. A practical question is not only whether adaptation helps, but how performance is likely to evolve as additional labeled target data become available.

The direct approach is to adapt the model at several target-data sizes and fit or extrapolate the resulting learning curve. This requires both repeated training and, in real applications, progressively larger labeled datasets. This study tests whether source-model properties, the model’s relationship to a small target sample, and limited early adaptation behaviour can instead be used to forecast the later curve.

Two settings are considered. *Cold start* uses only pre-adaptation information. *Warm start* additionally uses observations from the first two adaptation budgets. The aim is to demonstrate a complete experimental design for studying this forecasting problem rather than to establish a generally optimal predictor from this small PACS setup.

2 Related Work

PACS contains four visually distinct domains—art painting, cartoon, photo, and sketch—and was introduced as a domain-generalization benchmark [LYSH17]. Its strong distribution shifts make it useful for controlled adaptation experiments. The present study uses both a compact convolutional network and ResNet-18 [HZRS16].

Many domain-adaptation methods characterize or reduce source–target distribution differences. Maximum mean discrepancy (MMD), for example, measures distributional difference in a reproducing-kernel Hilbert space [GBR⁺12]. Here, MMD and related feature-space distances are used only as predictor inputs. Learning-curve extrapolation has also been studied for estimating downstream data requirements [MLA+22]. The present pilot

considers whether model-relative and data-relative signals can enable forecasting before any target learning-curve observations are available, or improve forecasting once a small number of target observations have been collected.

3 Dataset and Experimental Configuration

PACS contains seven classes across four domains. Each domain was split deterministically into 70% training, 10% validation, and 20% test partitions.

Source models were trained from scratch on every one-domain and two-domain source configuration using a compact three-layer CNN and a randomly initialized ResNet-18. Two random seeds produced 40 source checkpoints and 96 valid source–target cases in which the target domain was absent from the source domains.

Each source checkpoint was independently adapted using nested target-training subsets corresponding nominally to 5%, 10%, 25%, 50%, and 100% of the target training pool. Because subsets were constructed class by class, all fitting and evaluation use the exact case-specific sample counts rather than only the nominal percentages. Source training and adaptation ran for 100 epochs, retaining the checkpoint with highest validation accuracy. Together with zero-shot target-test performance, the five adaptation budgets provide six measured points per learning curve.

4 Methodology

4.1 Parametric Learning-Curve Targets

Each empirical target-test curve is summarized by

$$A(n) = A_0 + (a - A_0) [1 - (n + 1)^{-c}], \quad (1)$$

where A_0 is fixed to measured zero-shot target-test accuracy, $A_0 \leq a \leq 1$, and $c > 0$. Only a and c are fitted when constructing the target representation; Ridge predictors then predict the full parameter vector $[A_0, a, c]$.

Across all 96 cases, the fitted curves achieved mean MAE 0.0287 and mean RMSE 0.0358 relative to the six measured test points. Prediction quality is evaluated in *accuracy space*: predicted parameters are constrained to valid ranges, reconstructed at each case’s actual target sample counts, and compared with measured target-test accuracies.

4.2 Cold-Start Prediction

Cold start is evaluated before any target adaptation training. Predictor inputs include model architecture, source training-set size, the first target-subset size, source validation accuracy, and zero-shot target validation accuracy and loss.

Using the original source model and the 5% target subset, four representation discrepancies are also computed: RBF-MMD squared, centroid Euclidean distance, centroid cosine distance, and normalized covariance Frobenius distance. Four predictors are compared: mean curve, metadata-only Ridge, discrepancy-only Ridge, and metadata-plus-discrepancy Ridge. Zero-shot target *test* accuracy is never provided as an input.

4.3 Warm-Start Prediction

Warm start assumes that zero-shot evaluation and the 5% and 10% adaptation runs are already available. Inputs therefore add selected epoch, training accuracy, validation accuracy, and optimization-step count from both early runs. Target-test accuracy from these stages is not used as a predictor.

Representation discrepancies are measured in three ways at the 10% stage: before adaptation, after loading the 10%-adapted checkpoint, and as their signed change

$$\Delta D = D_{\text{adapted}} - D_{\text{source}}. \quad (2)$$

Warm-start Ridge variants combine metadata and early observations with source discrepancy, adapted discrepancy, discrepancy change, or adapted discrepancy plus change. Two simple baselines are also included: a mean remaining curve and a three-point logarithmic extrapolation fitted to zero-shot, 5%, and 10% target *validation* accuracies. Warm-start evaluation covers only the future 25%, 50%, and 100% budgets.

4.4 Evaluation Protocol

Generalization is evaluated with leave-one-target-domain-out outer validation: all cases targeting one PACS domain are held out while predictors are trained on the remaining three. Ridge regularization is selected by inner grouped cross-validation, grouping cases by source checkpoint. Cold-start and warm-start MAEs correspond to different forecasting tasks and are interpreted separately.

5 Results

5.1 Parametric Curve Fit

The curve family achieved mean fitting MAE 0.0287, median MAE 0.0284, maximum per-case MAE 0.0577, and mean RMSE 0.0358. The asymptotic parameter a reached its upper bound of 1.0 in 95 of 96 cases, so the fitted curves should be interpreted mainly as compact representations within the measured target-data range rather than reliable long-range asymptotes.

5.2 Cold Start

Table 1 summarizes cold-start performance. Metadata Ridge performed best overall at 0.0896 MAE, while adding the tested global discrepancy measures did not improve it.

Table 1: Cold-start MAE. Curve reconstruction uses the exact case-specific sample counts.

Model	0	5%	10%	25%	50%	100%	Overall
Metadata Ridge	0.0394	0.0849	0.0805	0.0994	0.1138	0.1194	0.0896
Metadata + discrepancy Ridge	0.0509	0.1054	0.1040	0.1228	0.1358	0.1421	0.1102
Mean curve	0.1167	0.1138	0.1115	0.1248	0.1301	0.1300	0.1212
Discrepancy Ridge	0.1089	0.1140	0.1142	0.1250	0.1367	0.1411	0.1233

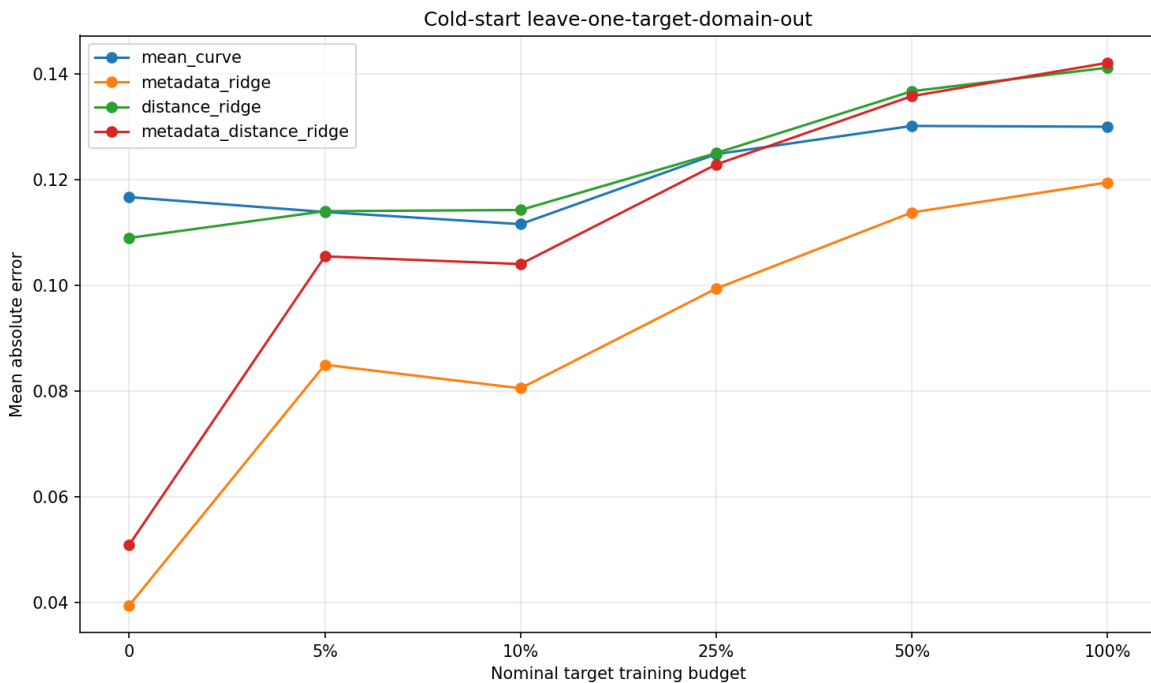


Figure 1: Cold-start mean absolute error across target-data budgets.

Metadata Ridge was best for three of the four held-out target domains; discrepancy-only Ridge was best for cartoon. Art painting was the most difficult cold-start target. Overall, source-model and zero-shot metadata car-

ried useful forecasting information, whereas the tested global representation discrepancies did not add consistent value.

5.3 Warm Start

Table 2 reports prediction of the future curve after observing the first two adaptation stages. Three-point logarithmic extrapolation was strongest overall at 0.0521 MAE, followed by metadata-plus-observation Ridge at 0.0600. Among discrepancy-augmented Ridge variants, signed discrepancy change performed best at 0.0686, but still did not improve on metadata plus early observations.

Table 2: Warm-start MAE for the future adaptation budgets.

Model	25%	50%	100%	Overall
Three-point log curve	0.0398	0.0500	0.0664	0.0521
Metadata + observation Ridge	0.0509	0.0616	0.0675	0.0600
Metadata + observation + change Ridge	0.0574	0.0705	0.0778	0.0686
Observation Ridge	0.0603	0.0724	0.0805	0.0711
Metadata + observation + source discrepancy Ridge	0.0597	0.0770	0.0864	0.0744
Metadata + observation + adapted discrepancy Ridge	0.0629	0.0783	0.0869	0.0760
Metadata + observation + adapted discrepancy + change Ridge	0.0677	0.0842	0.0923	0.0814
Mean remaining curve	0.1248	0.1301	0.1300	0.1283

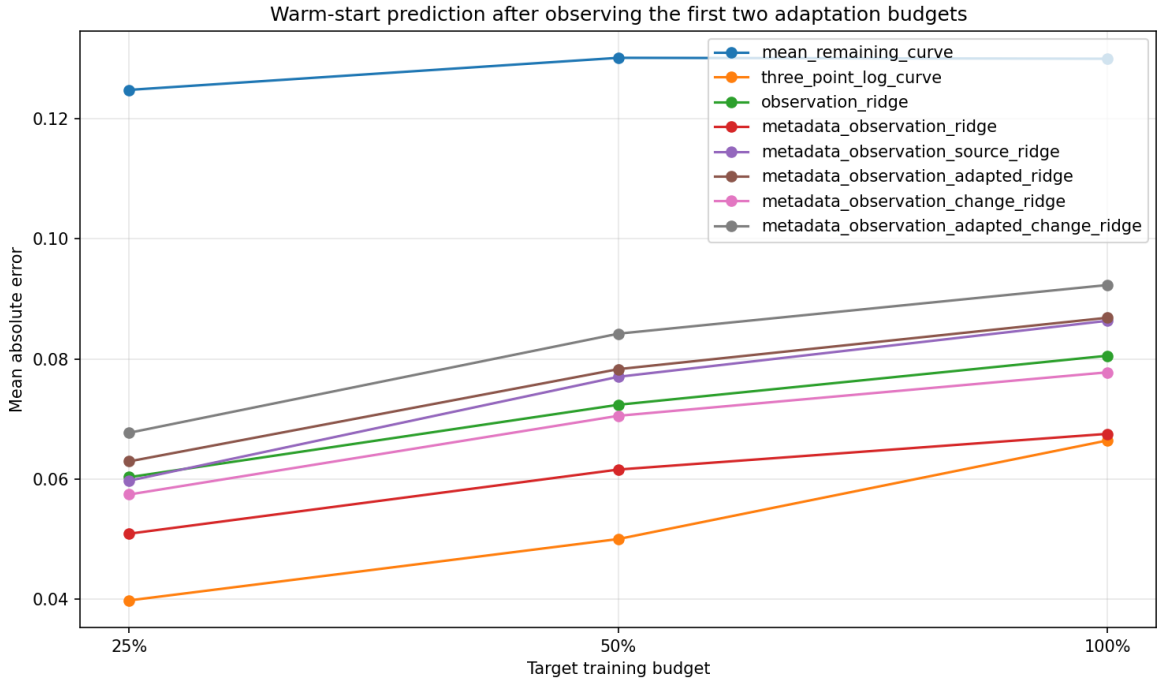


Figure 2: Warm-start mean absolute error after observing the 5% and 10% adaptation stages.

The best method varied by held-out domain, but the aggregate result is simple: the early validation trajectory itself was highly informative, while the tested discrepancy summaries did not improve the strongest warm-start predictor.

5.4 Interpretability and Ablation

The interpretability analysis treats standardized Ridge coefficients and ablations as descriptive properties of the fitted predictors rather than causal feature importance. In cold start, the small metadata feature set supports individual leave-one-feature-out analysis. Zero-shot target validation accuracy had the largest mean absolute

standardized coefficient (0.4177), and removing it increased cold-start MAE from 0.0896 to 0.1266. No other single cold-start removal increased MAE by more than 0.0069.

Warm-start inputs are substantially more redundant. In particular, n_5 and n_{10} had correlation 0.99999; n_5 and the 10% optimization-step count had correlation 0.9943; and the 5% and 10% validation accuracies had correlation 0.9707. Because such redundancy can make individual coefficients and single-feature removals unstable or misleading, warm-start ablation is performed on coherent information families rather than one feature at a time. Results are shown in Table 3.

Table 3: Grouped warm-start ablation. Positive Δ MAE indicates that removing the group worsened prediction.

Removed feature group	MAE	Δ MAE
None (full warm Ridge)	0.0600	0.0000
Early validation performance	0.0827	+0.0227
Source / zero-shot context	0.0686	+0.0086
Budget / training amount	0.0562	-0.0038
Training dynamics	0.0550	-0.0050

The clearest warm-start signal came from the 5% and 10% validation accuracies: removing that group increased MAE from 0.0600 to 0.0827. Source and zero-shot context also helped, although more modestly. In contrast, removing the budget/training-amount or training-dynamics groups slightly improved held-out performance, consistent with those strongly correlated variables adding redundancy or noise rather than unique predictive information. The grouped analysis is therefore the primary warm-start ablation diagnostic.

6 Discussion and Limitations

The cold-start results suggest that source-model and zero-shot metadata can carry useful information about later adaptation performance in this setup. Cold-start LOFO analysis particularly highlighted zero-shot target validation accuracy, while the tested global representation discrepancies did not improve the strongest metadata model. In warm start, early adaptation behaviour was more informative: a simple extrapolation of the first three validation observations performed best overall, and grouped ablation confirmed that the early validation accuracies were the most useful feature family inside the Ridge predictor.

The discrepancy-change variant performed better than static source or adapted discrepancy alone, but remained weaker than metadata plus observations. This makes it a potentially interesting signal to investigate rather than evidence of a general relationship. Likewise, the grouped ablation results are best read as diagnostics of this pilot configuration, especially given the small number of target domains and the strong correlations among several inputs.

The parametric target is another limitation. Equation 1 fits the measured observations well, but the fitted asymptote reaches its upper bound in 95 of 96 cases. The measured budgets therefore constrain the local curve shape much better than the true long-run asymptote.

Finally, the experiment is limited to four PACS domains, two architectures, two seeds, one supervised adaptation procedure, and linear Ridge predictors. In a full PhD study, the same framework could be tested across broader model, dataset, and adaptation families and with richer structural or class-conditional signals.

7 Conclusion

This pilot constructs an end-to-end framework for forecasting domain-adaptation learning curves from information available before and shortly after adaptation begins. Across 96 PACS source–target cases, metadata Ridge achieved 0.0896 MAE in cold start, while three-point logarithmic extrapolation and metadata-plus-observation Ridge achieved 0.0521 and 0.0600 respectively in warm start.

The most useful practical observation from the pilot is that early validation behaviour carried substantially more forecasting information than the tested global representation-discrepancy summaries. Grouped ablation supported the same interpretation: removing early validation performance caused the largest degradation, while removing redundant budget and training-dynamics variables slightly improved generalization. The experiment therefore serves as a concrete example of how the proposed PhD question—predicting future adaptation performance and data requirements from limited early information—can be formulated, evaluated, and interpreted experimentally.

References

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